From downloads to discussions: Leveraging Yet Another Positioning Solver (YAPS) and modern movement analysis to transform raw telemetry data into interpretable results





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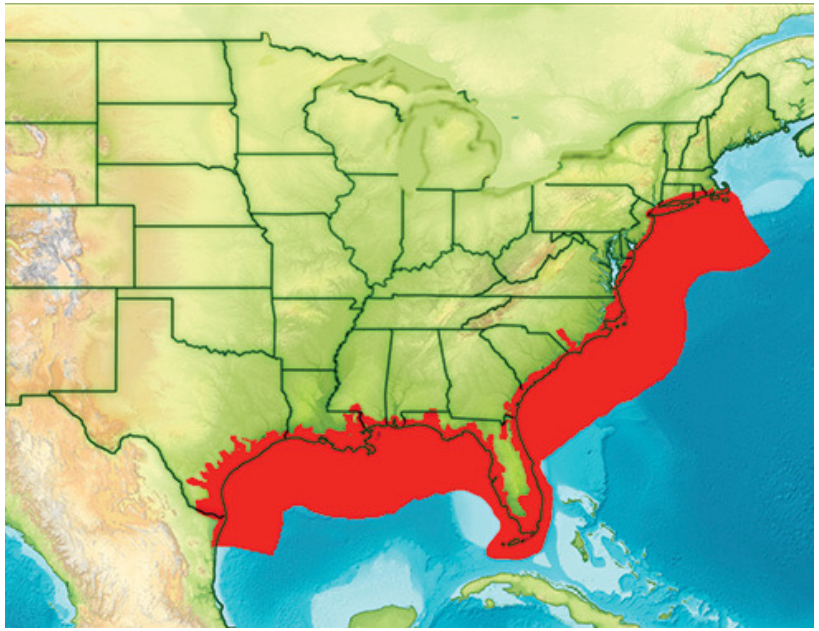
[2]Department of Earth and Environment, Florida International University, Miami, FL, US

Outline for today

1. Scales of Habitat Selection
2. Modeling Habitat Selection – acoustic telemetry
3. Fine-scale habitat selection
4. Considerations and Decision Breakpoints
5. Hands-on example with code using Spotted Seatrout data

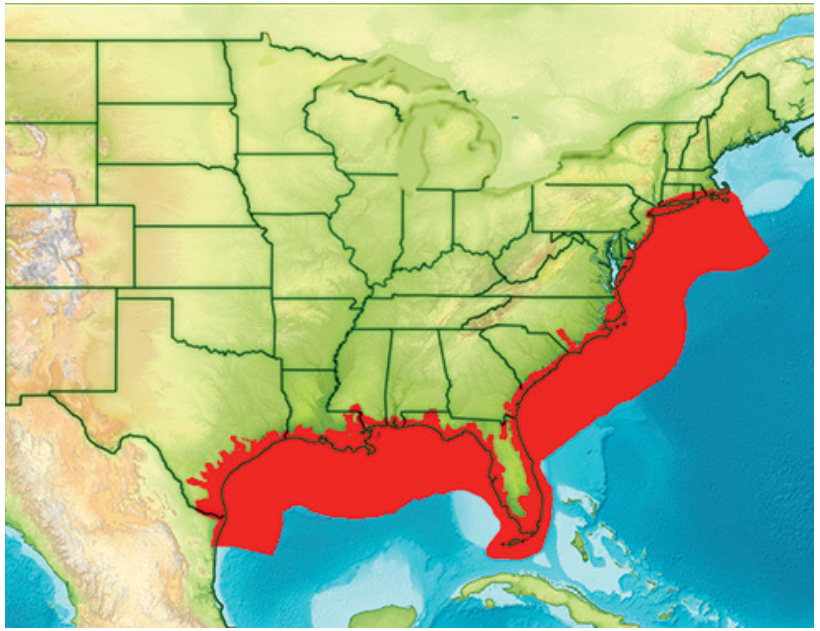
Habitat selection occurs on multiple scales

First-order: Ecological Range

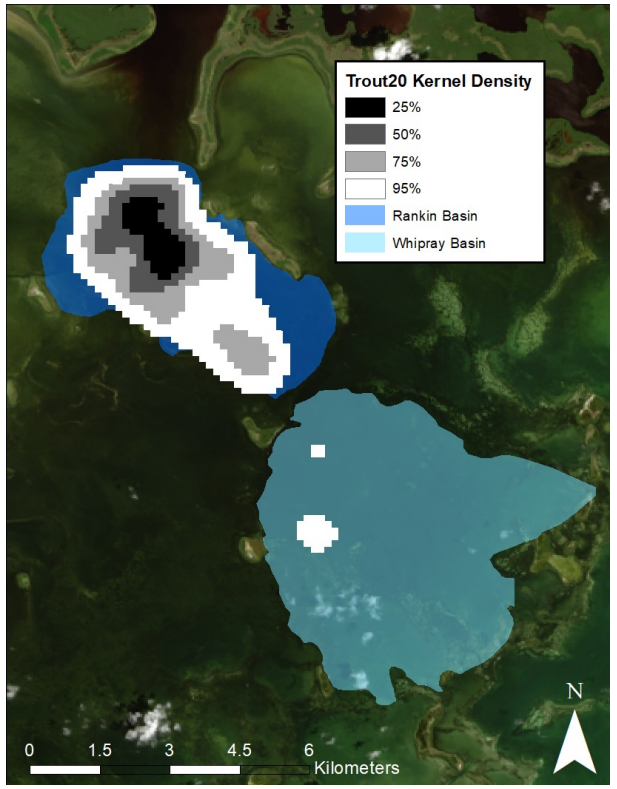


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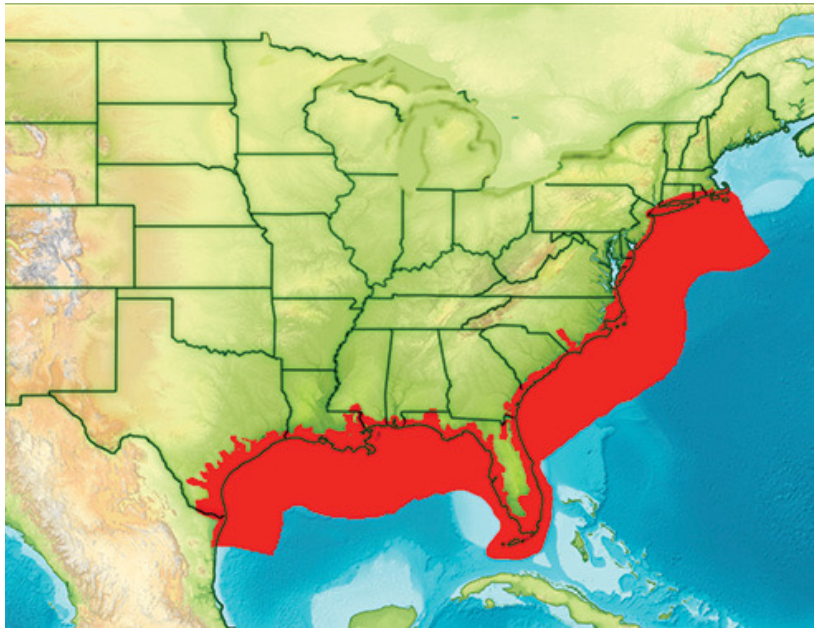


Second-order: Home Range

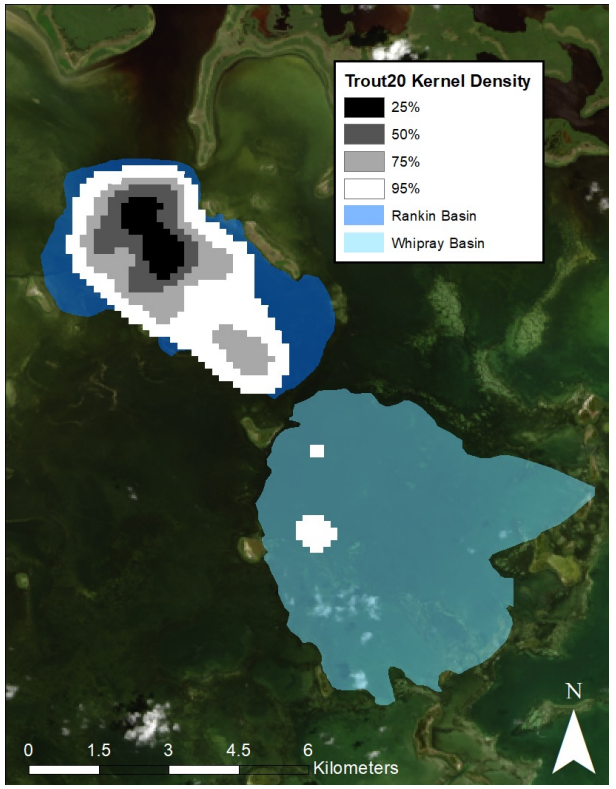


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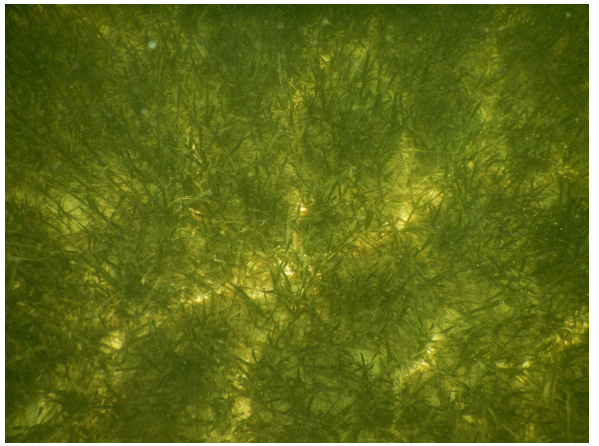
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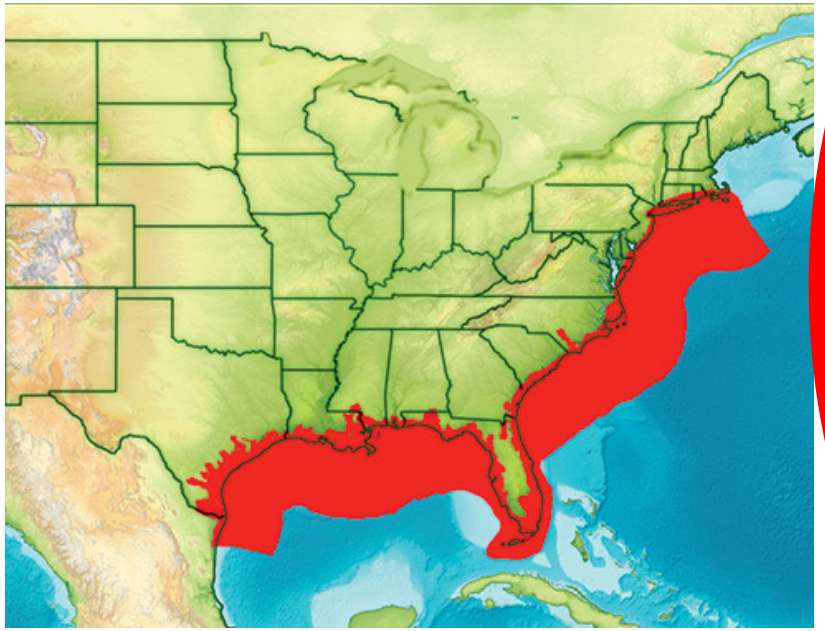
Third-order: Habitat Patches



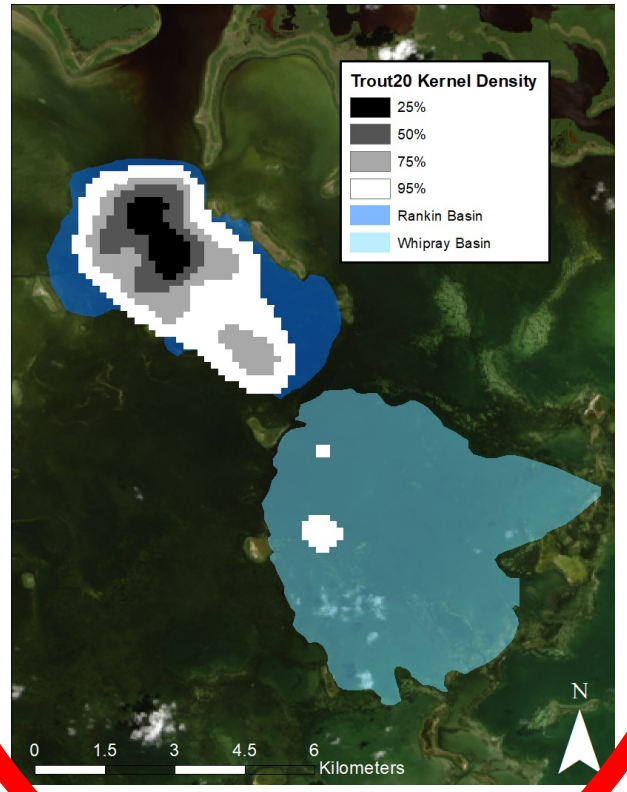


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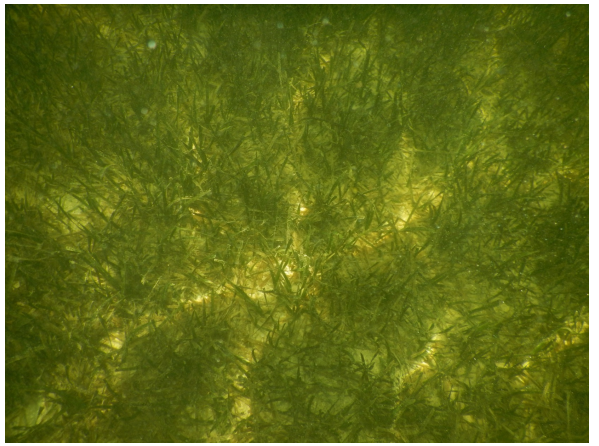
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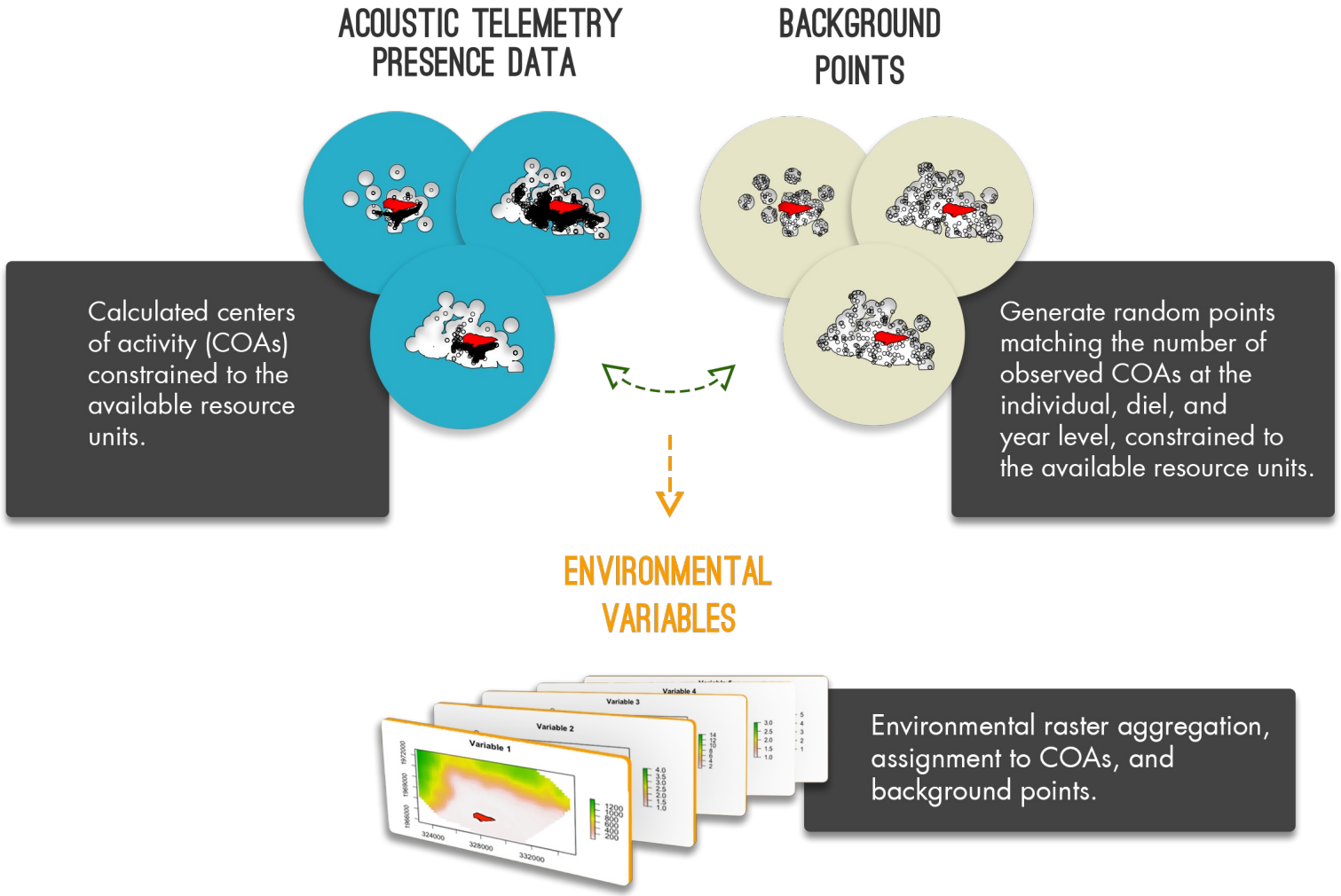


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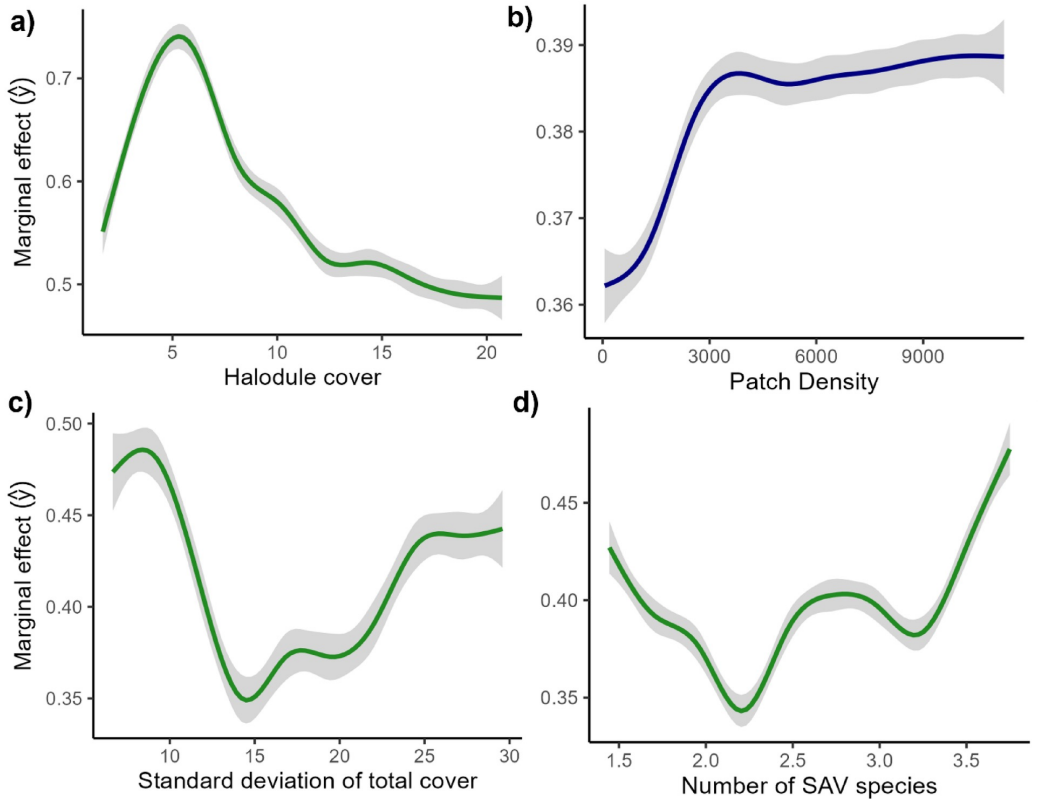
Second-order Selection: Resource Selection Functions



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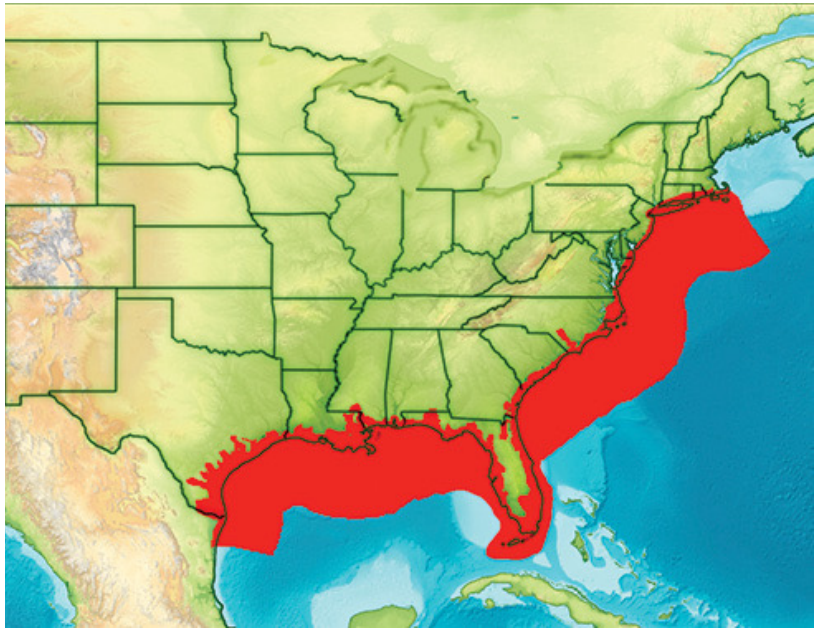
Relative Habitat Selection and Resource Selection Functions in Aquatic Acoustic Telemetry: Theory, Application, and Process

Hierarchical habitat selection by a predatory fish in a patchy seascape

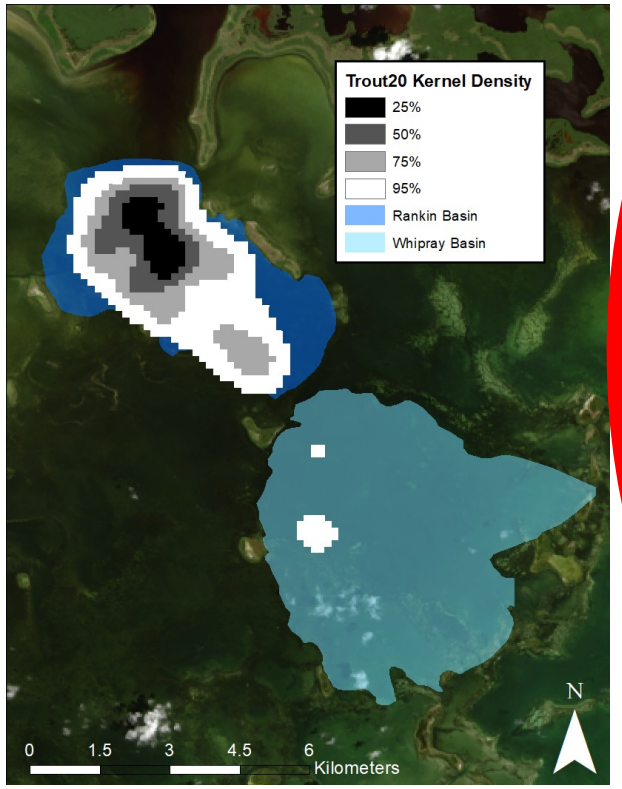


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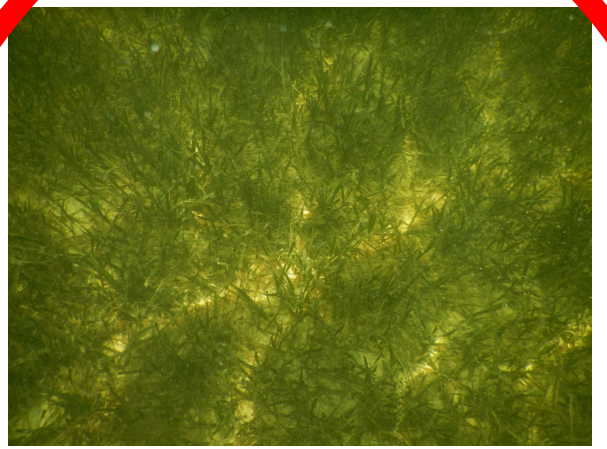
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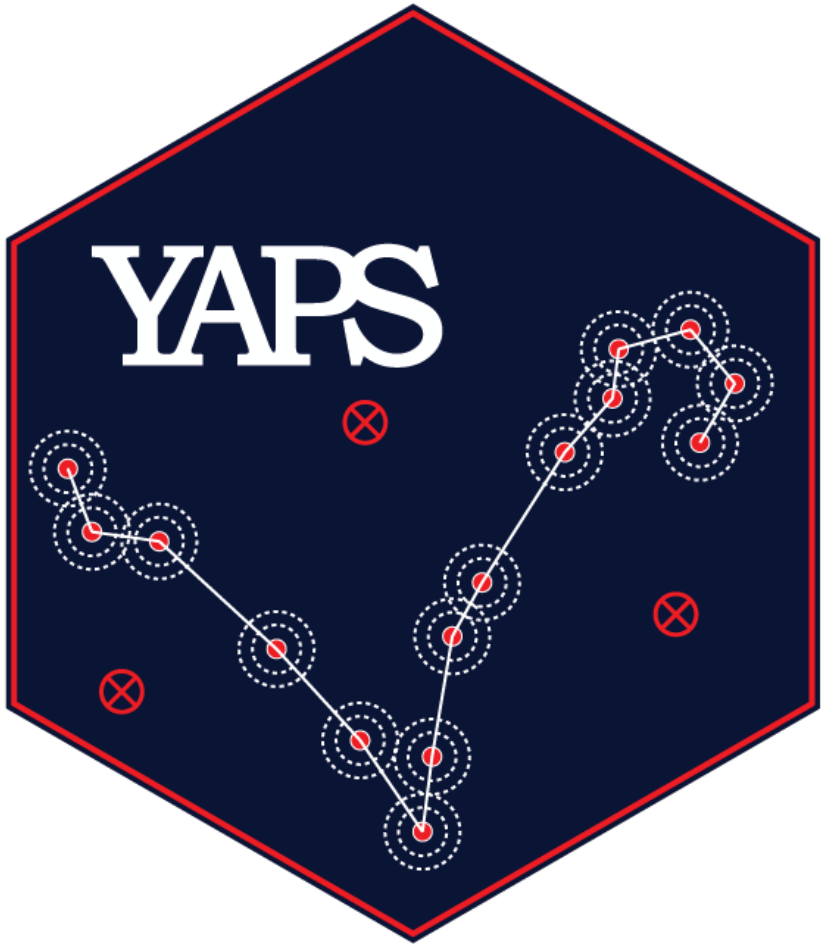


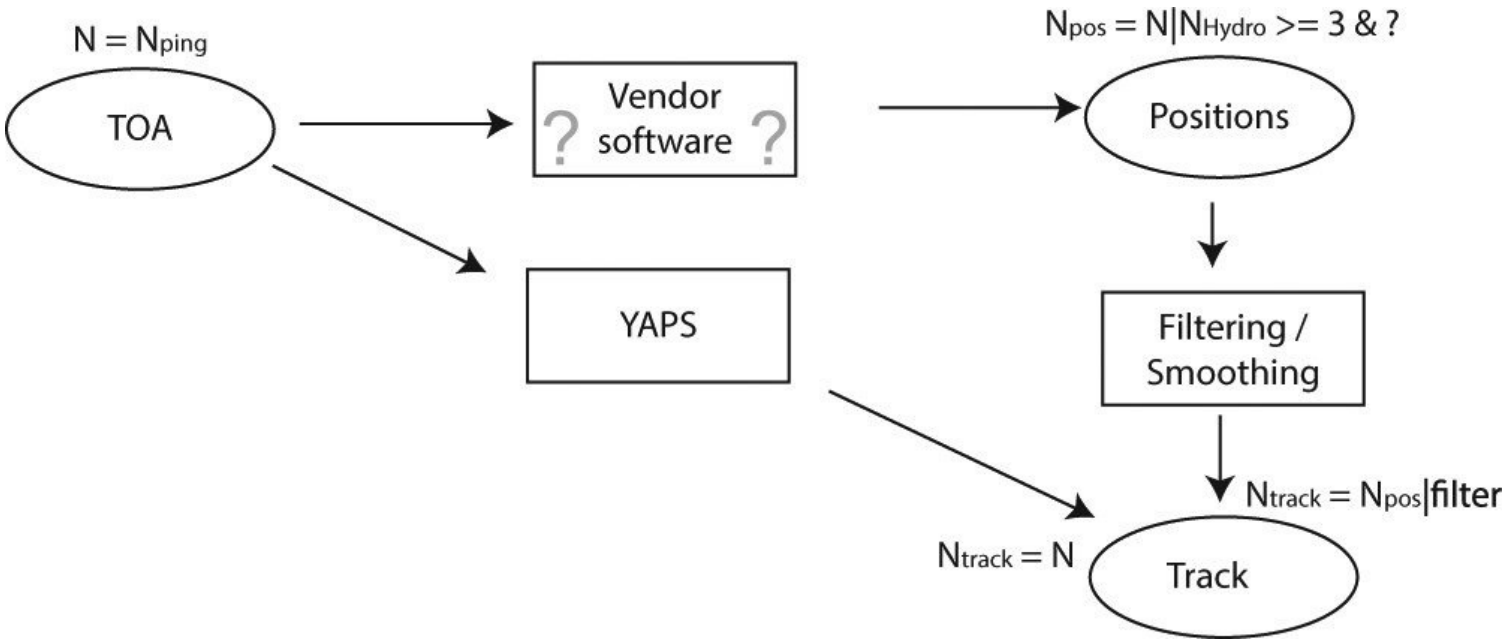
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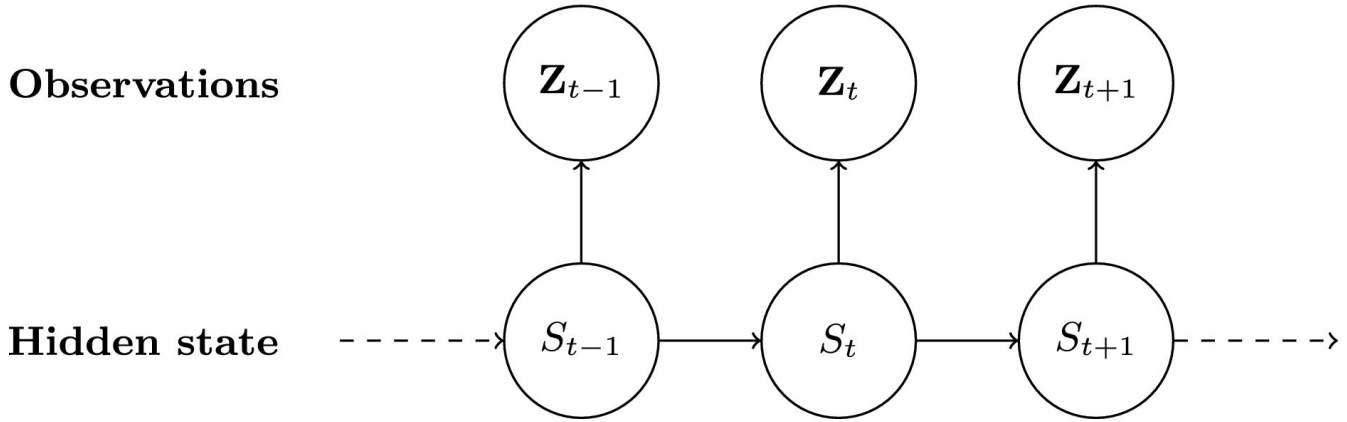


Third-order Selection: Yet Another Positioning Solver (YAPS)

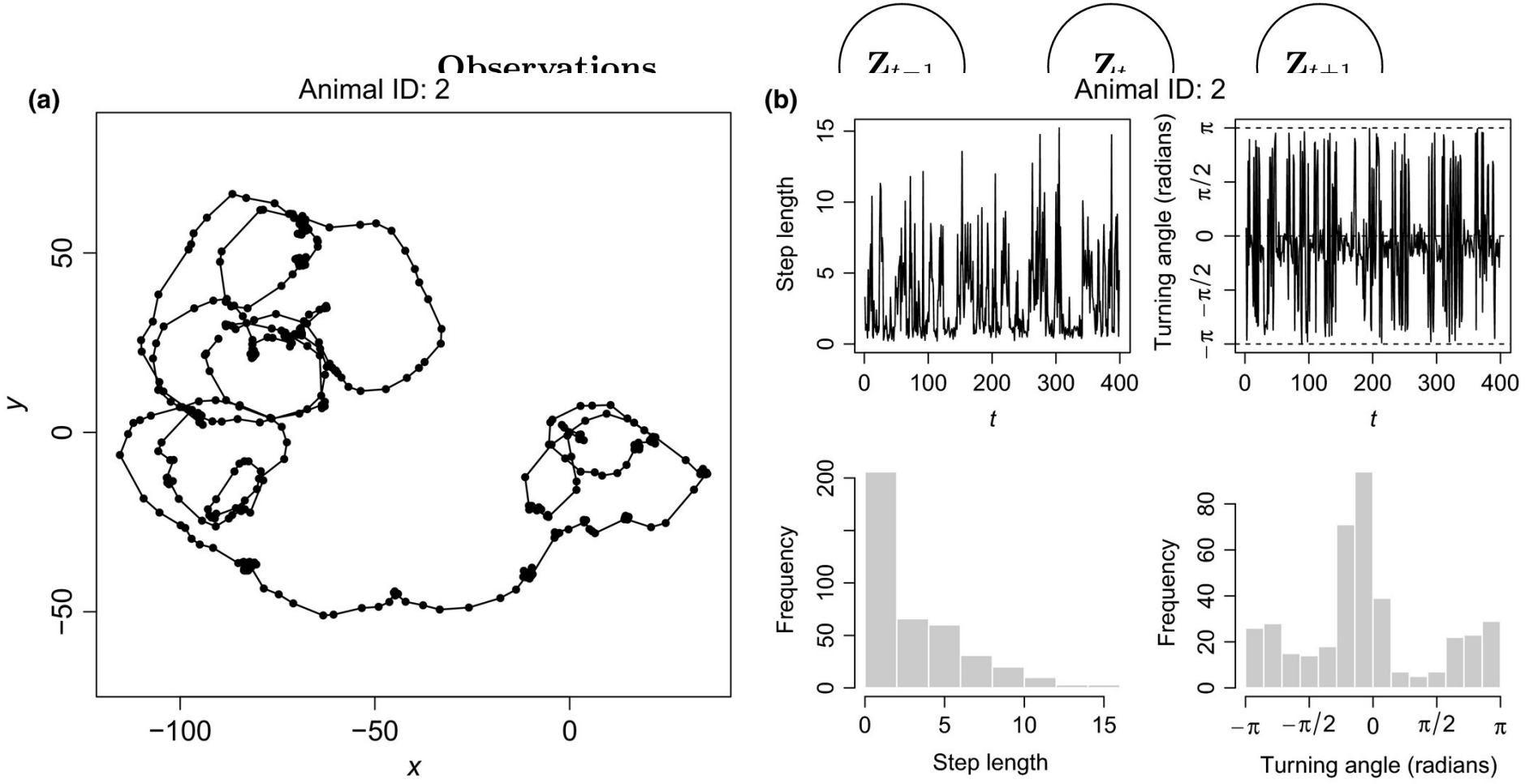




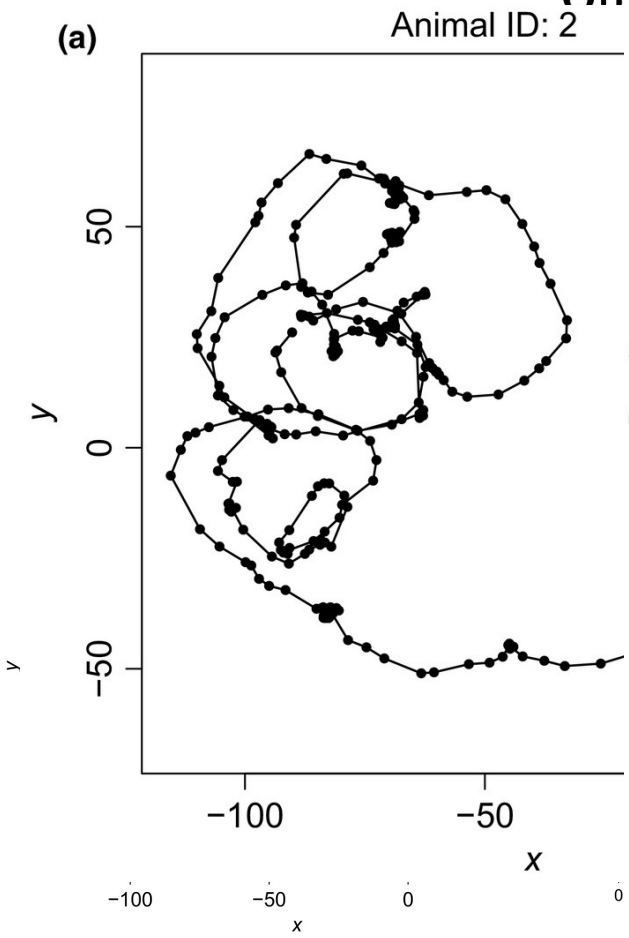
Third-order Selection: Hidden Markov Models (HMMs)

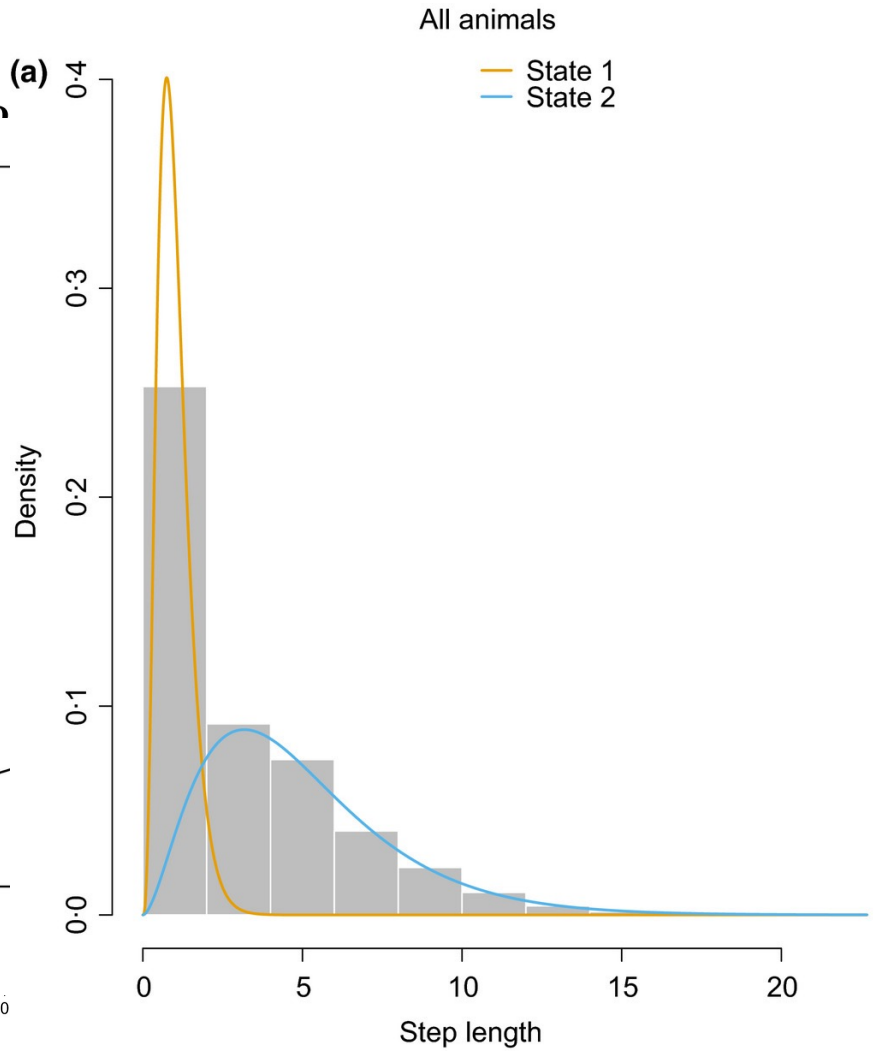


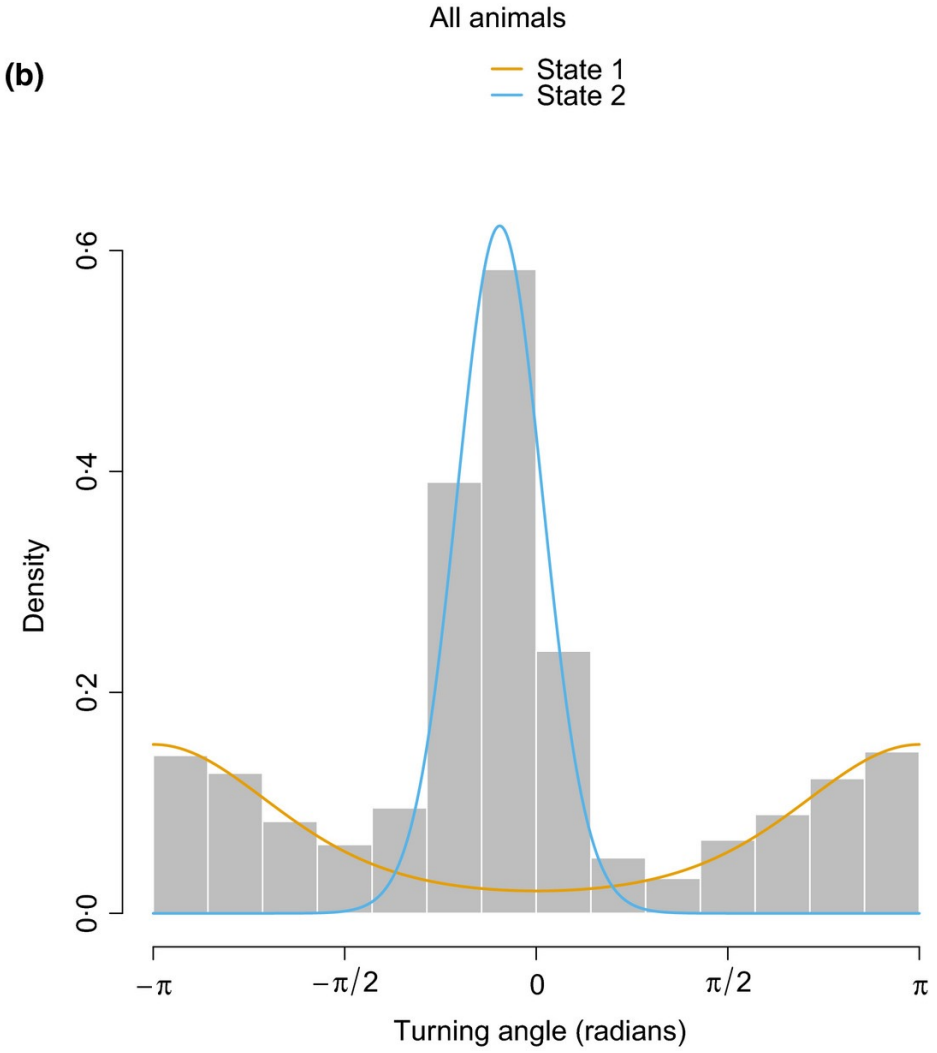
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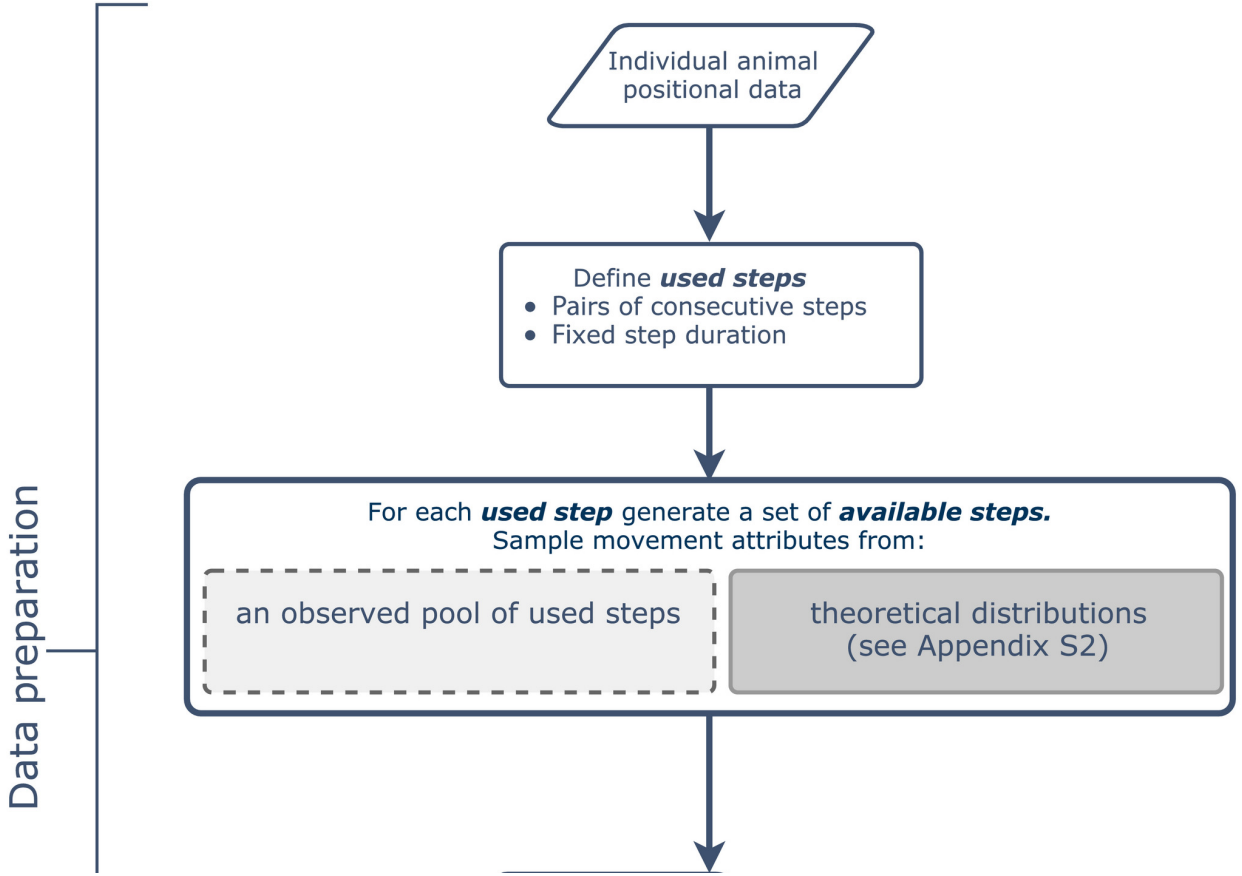
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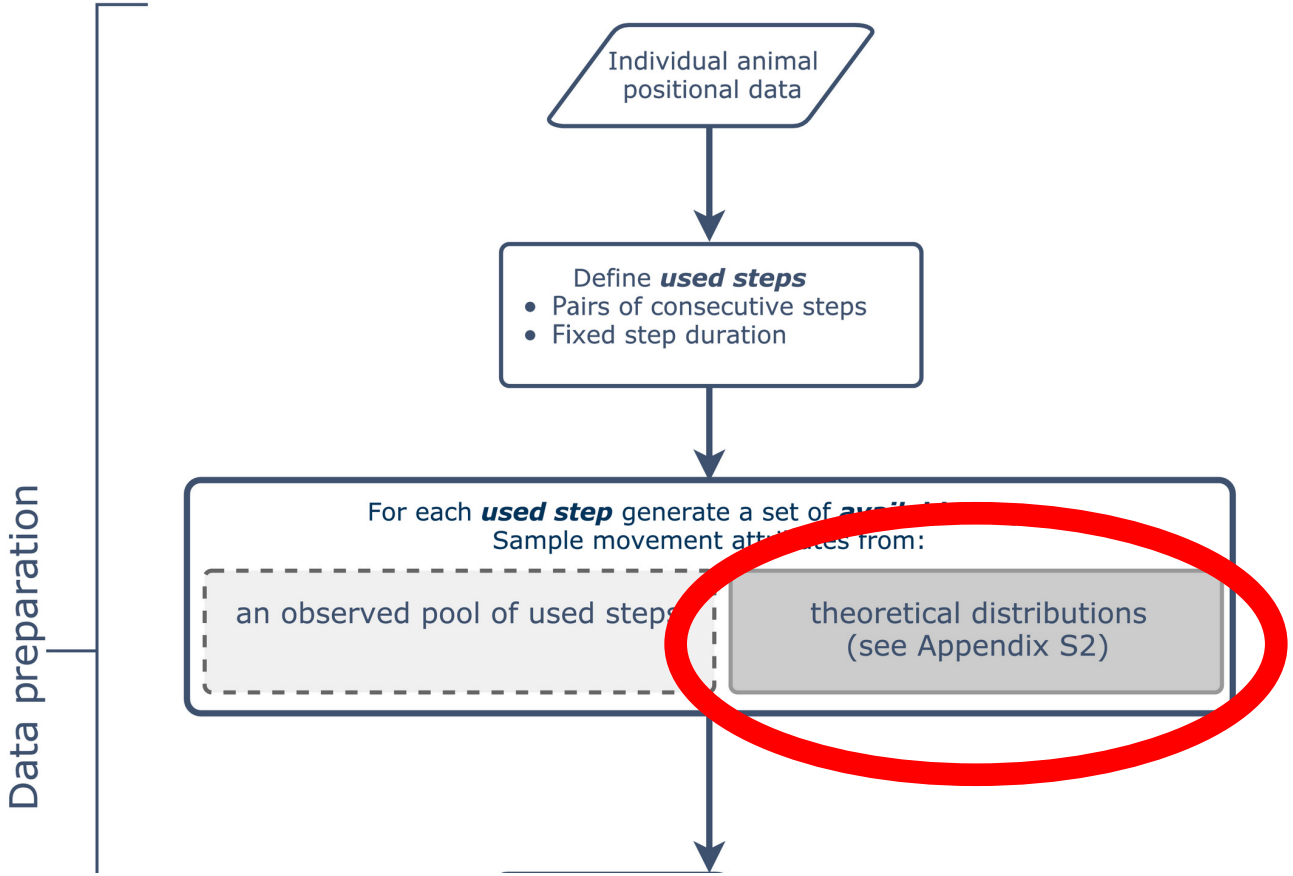




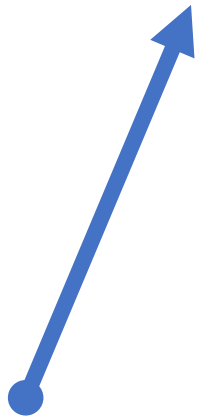
Third-order Selection: integrated Step Selection Functions (iSSFs)



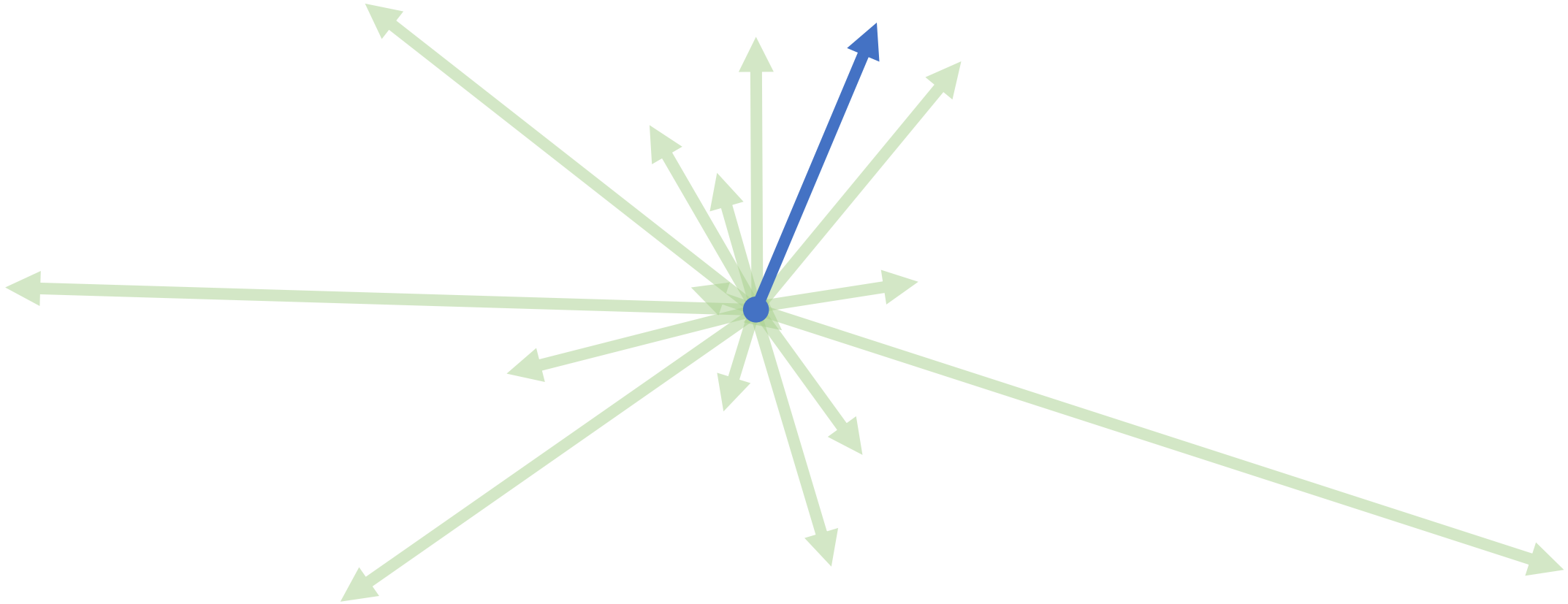
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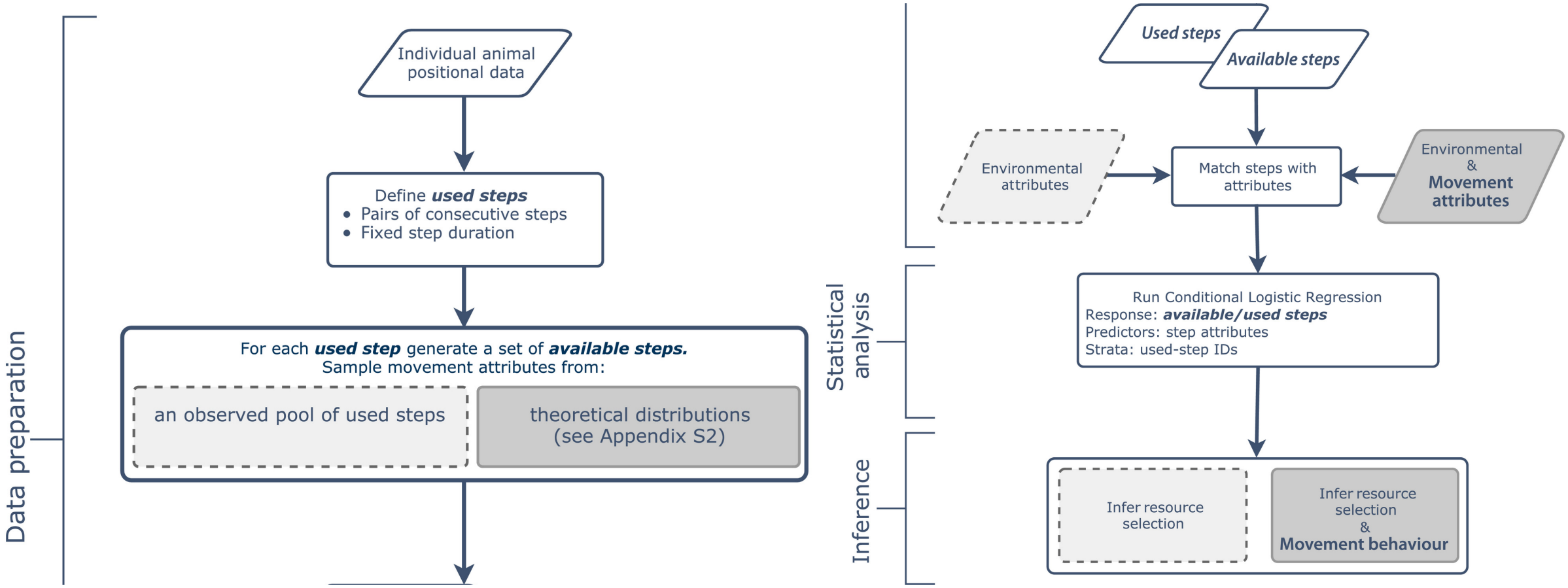
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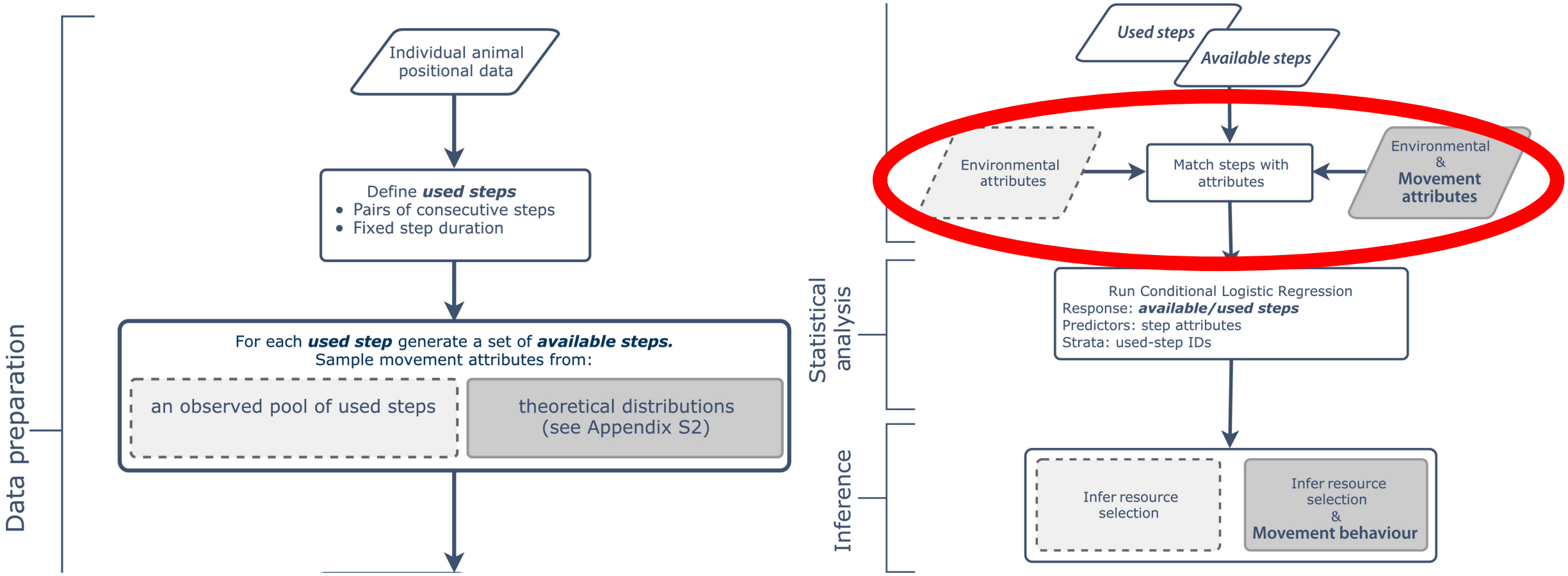
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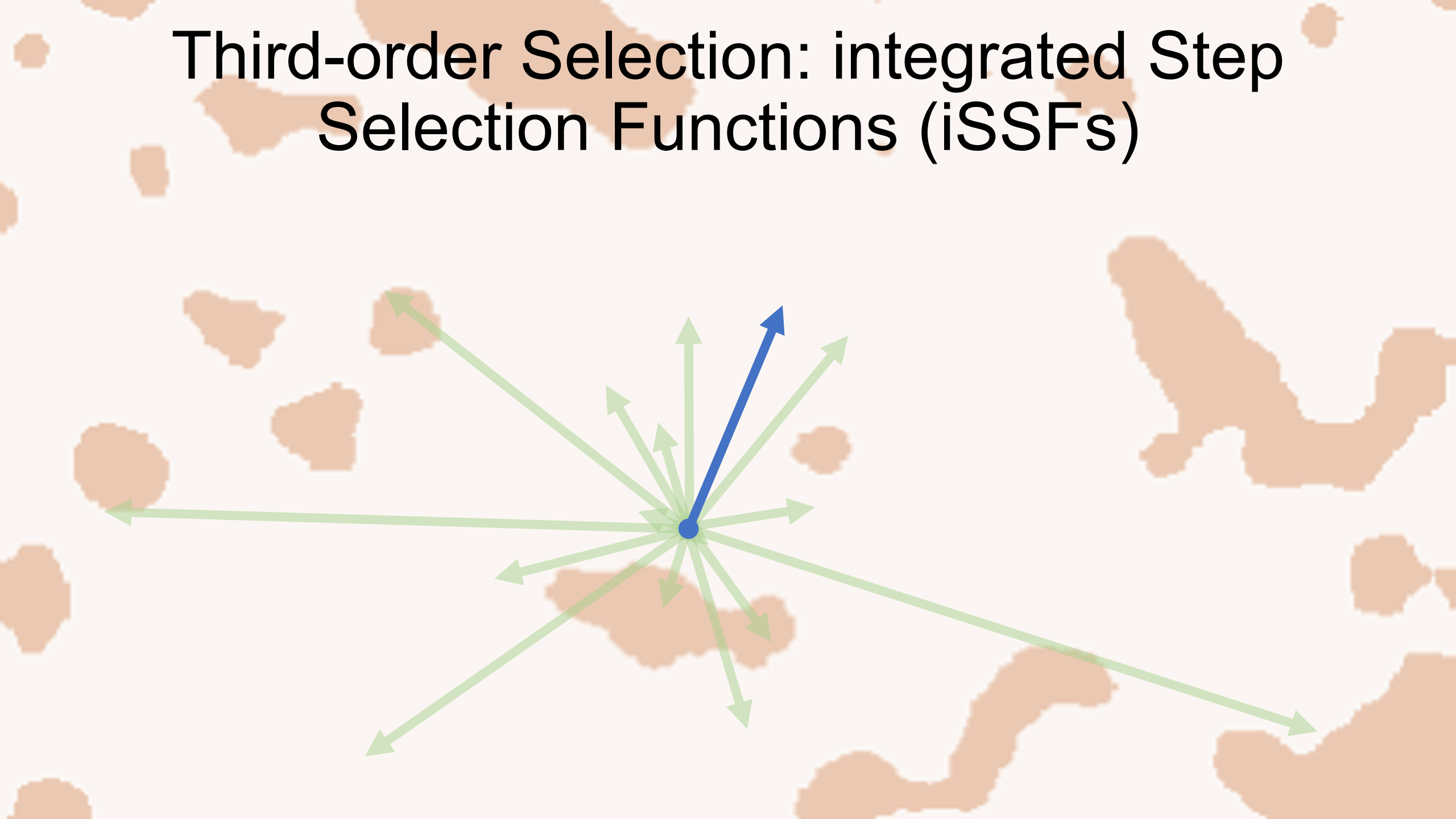
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Considerations and Decisions in YAPS, HMMs and iSSFs

Array design

- Is it an array designed for fine-scale positioning?
- Does it cover enough habitat variation?

Tag specifications

- What is the time delay of your tags? This will impact the timing of your steps

Accuracy of calculated positions

- What is the allowable accuracy of your calculated positions? This will impact the resolution of your environmental variables

Track specifications

- How long do you wait between positions before you start a new track? What constitutes a fish leaving the array?
- What tracks are worth keeping? How long do they need to be?

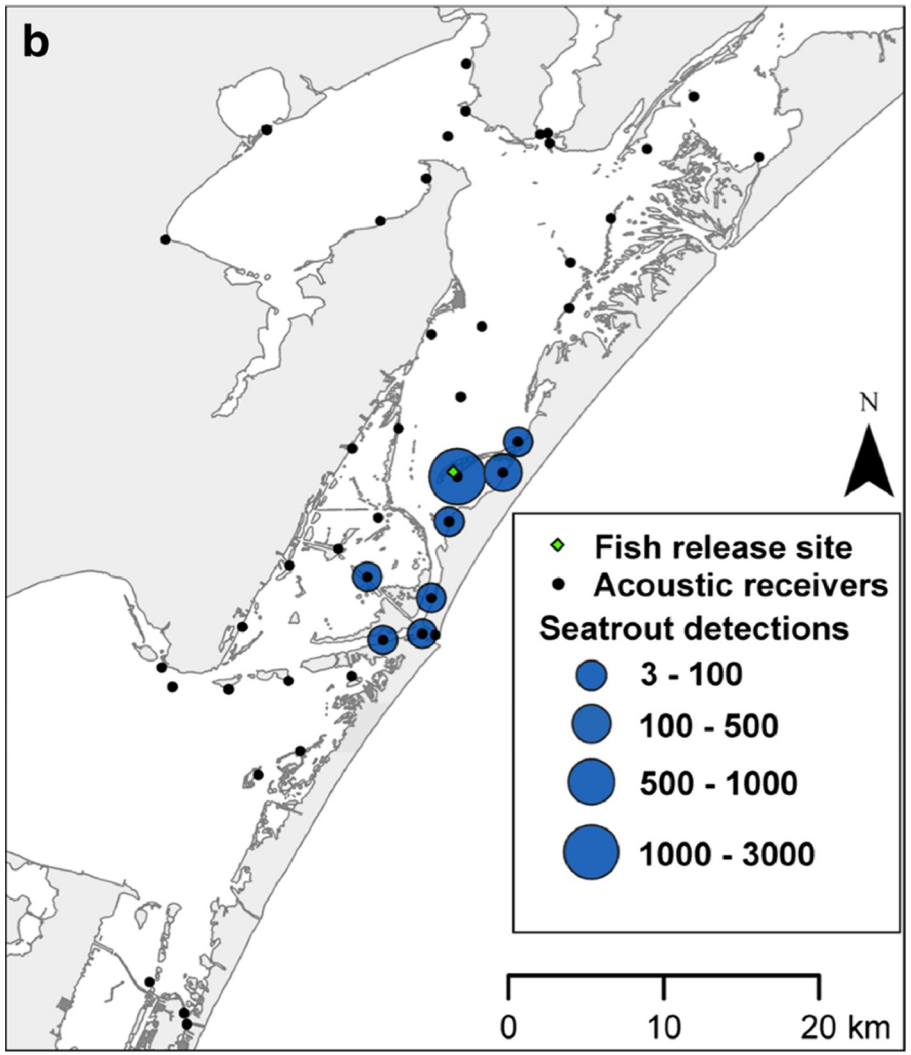
Environmental variables

- Keep in mind the resolution of your variables! Also are they static or dynamic?

Seatrout Example

Spotted Seatrout (*Cynoscion nebulosus*)

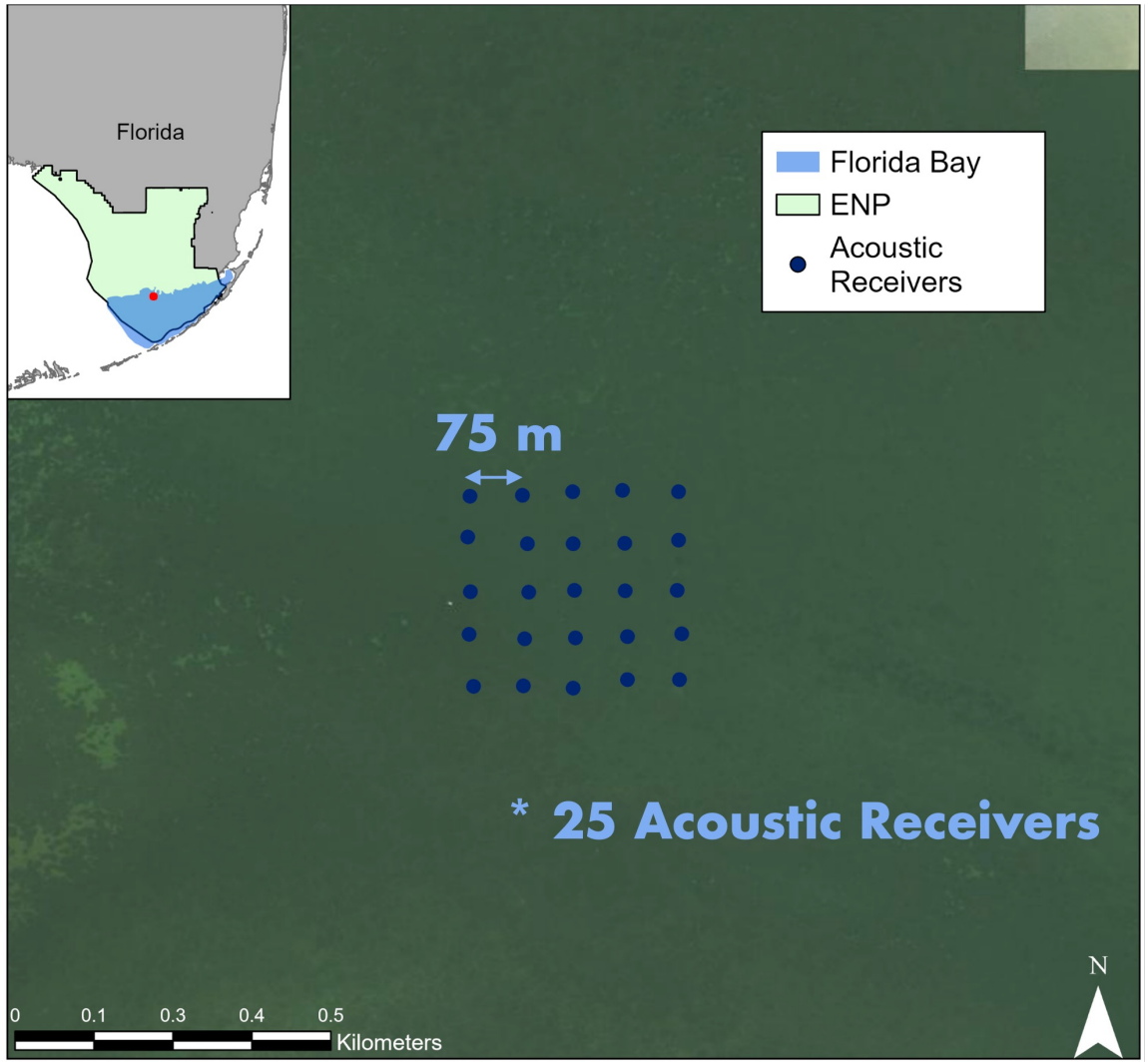






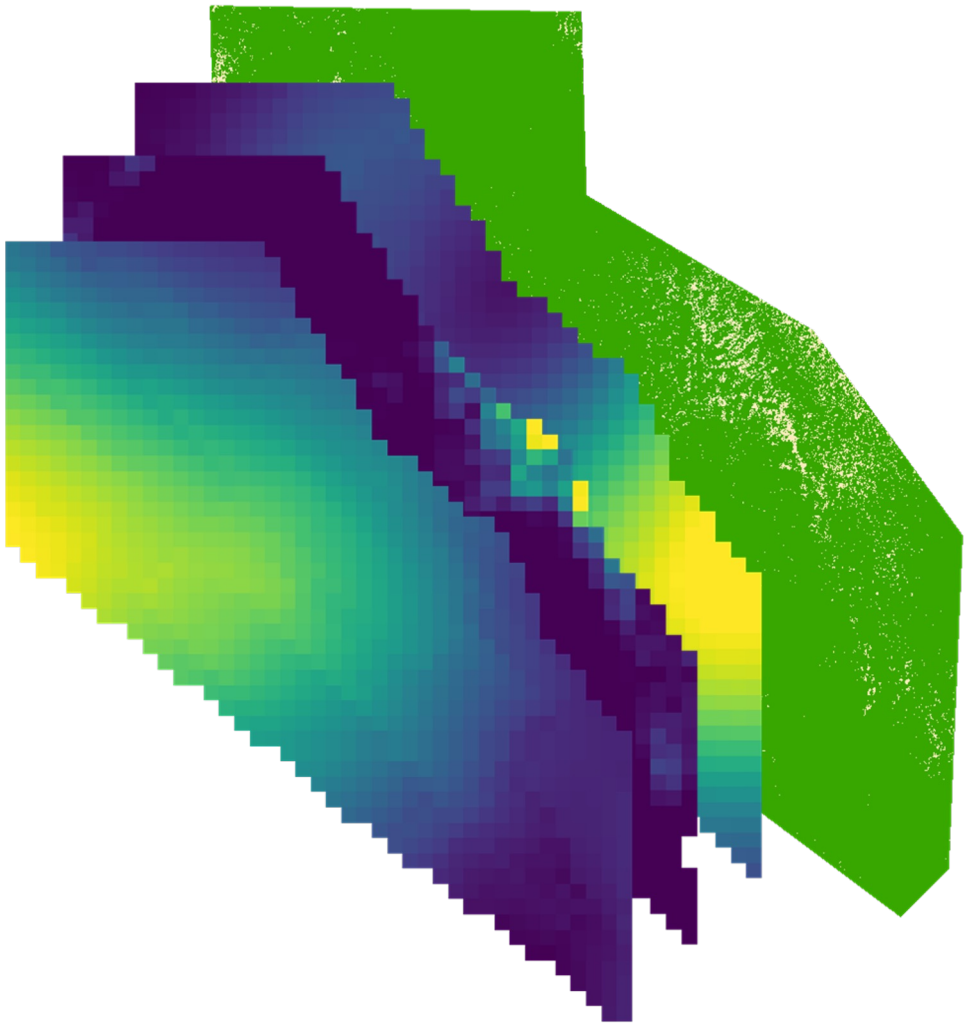
Moulton et al., 2017 *Estuar Coasts*

Methods



- 25 receivers
- 151 tagged seatrout
  - Using a subset of 3 individuals

Methods



SAV surveys
Aerial imagery
Spatial Pattern Metrics

Let’s move on over to the script!